

Molecular Lab #8

Restriction endonuclease sequences

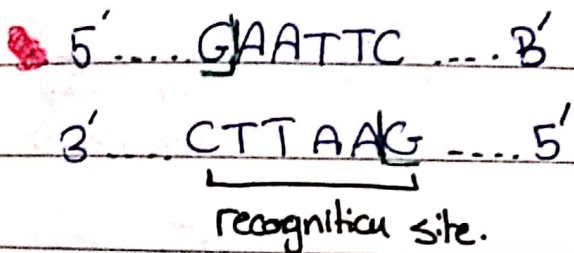
A protein that recognizes a particular sequence of DNA and cuts the DNA at that site (the restriction site).

Endonuclease within sequence, exonuclease على الأطراف

1. Source: Bacteria.

EcoRI → roman number
 ↓ ↓ ↓
 gene strain
 ↓
 species
 ↓
 from E.col.

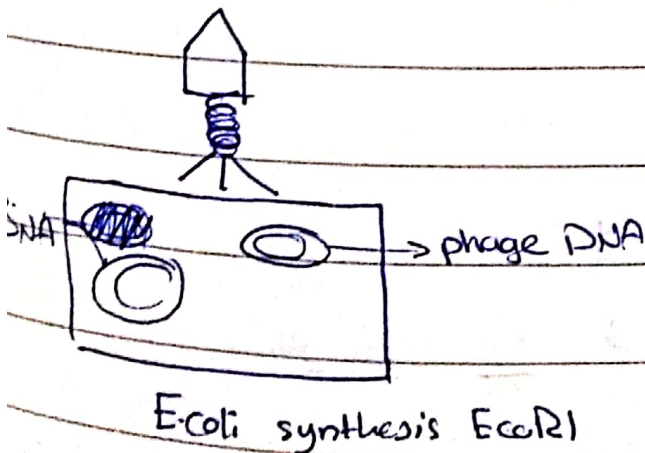
* At least each restriction enzyme have 3 letters & 1 number
4 → strain, 5 → substrain (capital)



→ The cut happens on dsDNA
not ssDNA

* Cut phosphodiester bond.

3. Restriction enzyme $\rightarrow$ Immune system for bacteria
(Defense mechanism)



Ecok1 enters virus cells and scan every recognition sites cut & cleavage. (and there is more than one R.S) & cleavage Fragmentation.

modification system. ؟

(Methylation on C, A)

4. Restriction methylation modification system.

How the bacterial cell protect itself from Restriction Enzyme.

Methylation: A: O6 → mask H_2O

C: C5, N4

5. Recognition site = palindromic sequence.

↳ Same reading same direction
in opposite strand.

Type 1

Type 3

| rs |
cut $\approx 1000\text{Kb}$
↓
(non-specific)

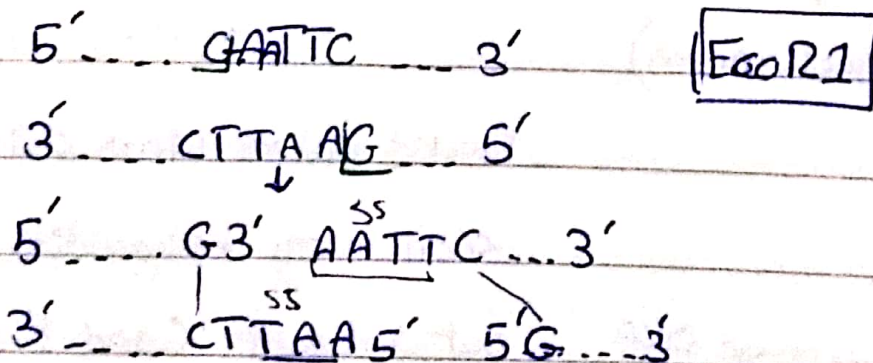
| rs |
Cut $\approx 25\text{bp}$.
non specific.

Randomly.

ATP مطلوب

- We use type 2 (commercially available)

- Types of ends



sticky 5' over hang. $\leadsto$ ss الأسبفون

Pst1

5' ... CTG CAG ... 3'

3' ... GAC GTC ... 5'

5' ... CTG CA 3' 5' G ... 3'

3' ... G 5' 3' AC GTC ... 5'

sticky 3' over hang.

3' over hang ← 3' بقعة من جهة 3'

(انتر) 5' over hang ← 5' بقعة من جهة 5'

5' ... CCG GGG ... 3'

3' ... GGG CCC ... 5'

5' ... CCC 3' 5' GGG ... 3'

3' ... GGG 5' 3' CCC ... 5'

Blunt Ends

Restriction Fragment length polymorphism (RFLP)

5' ATGCTAGGTA GCTATCTGATGACCCGGG TGA OCTAGGC 3'

250bp

50bp

TGAGTA CCTAGCT 3'

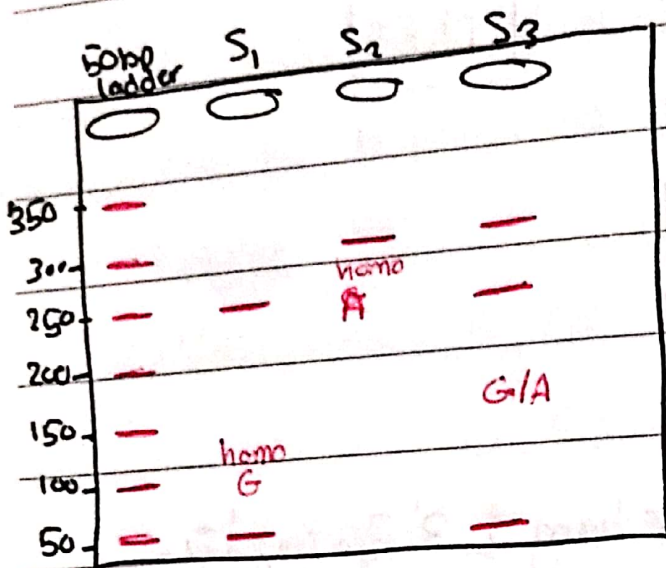
PCR product: 300bp

RFLP PCR product

Sma1 5' CCCGGG 3'

Cut (G) = 250 + 50

no cut (A) = 300



Mutation → Mutant

SRP → variant

Homo A → homozygous variant

3% Agarose gel

5' AGTTGC TGA CTGA CTGAGG AAG CCC GG GCTA GGAG

100bp

150 bp

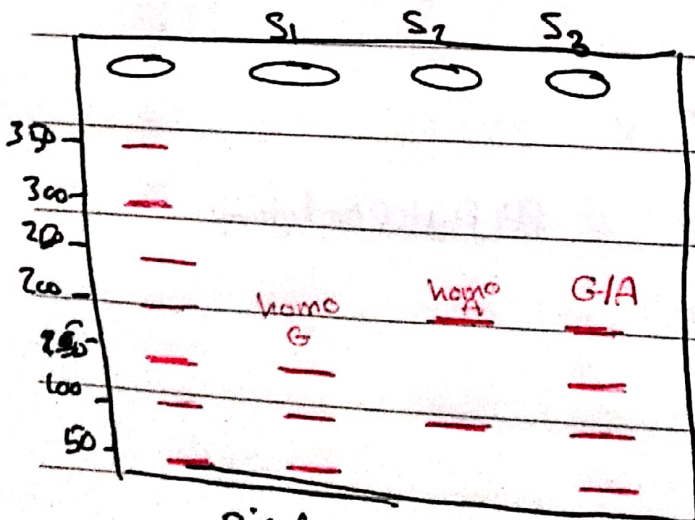
GGAGGTA CCC GGG CTA GTCT AG CTAG 3'

50bp

SmaI

Cut(G): 100, 150, 50

no Cut(A): 100, 200



3% Agarose gel

Ode1 5' - CTNAG - 3'
 Nucleotide.
 Five cutter

PRD1 → exon 4

PCR = 267 bp.

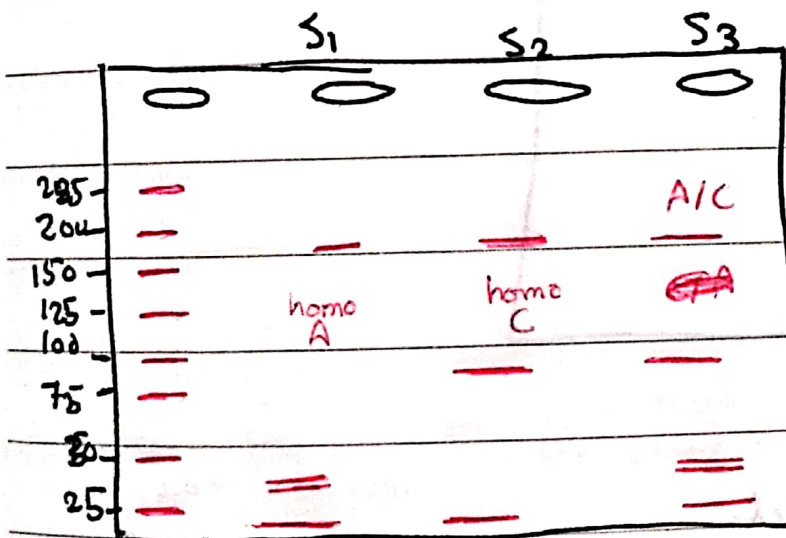
nt 1, 2, 3, 4, 5

5' GTACCTA CTAAG ATGGACTGTA CTAAG AG
 167 bp 44 bp 40 bp

TGAGAGCATCTGAGCTAG 3'
 16 bp

cut (A): 167, 44, 40, 16

no cut (C): 167, 84, 16.



4% Agarose gel

Low resolution in gel
 so (40, 44) will appear
 in one band

- RFLP rxn:- (one sample)

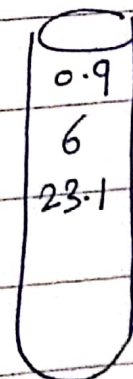
R.E ~~0.1-0.5~~ (0.1-0.5) M → 0.3 M → X3

R.E buffer (2 M) → X3

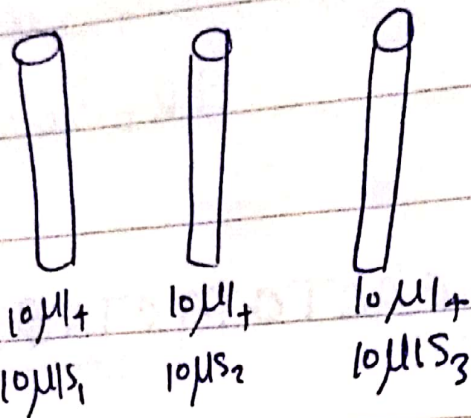
M.F.W 7.7 M → X3

PCR product 10 M ~~10 M~~

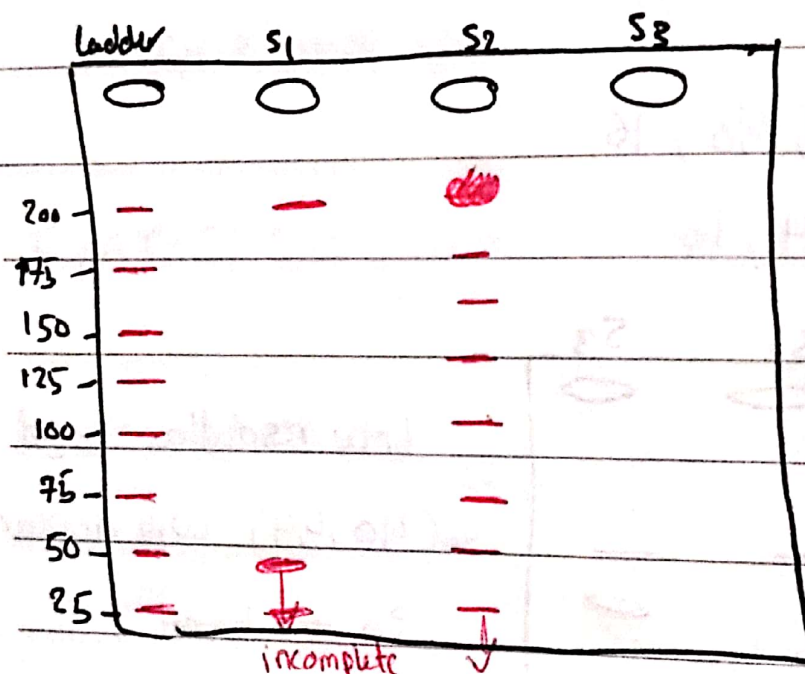
20 M



Vortex



→ incubate 37°C overnight.
 Optimum temperature for
 enzyme to work (cut).



No contamination
 in RFLP

(increase incubation time) → digestion

over digestion (more time) →

incubation time

4i. Agarose gel.